

1. Input Data & Normalization

Raw KT Logs (ASSISTments 2012, Junyi, XES3G5M) → Schema Standardization

2. Splitting & Stratification

Learner-based / Temporal Splits → Train-only KC-frequency Buckets

3. Baseline Evaluation

Model Training (IRT, DKT, SimpleKT) → Raw Prediction CSV Export

4. Diagnostic Metrics & Stratification Suite

- Aggregate Metrics: AUC, Accuracy, NLL, RMSE
- Calibration Strata: Expected Calibration Error (ECE) & Brier Decomposition
 - Visual Analytics: Multi-seed Reliability Diagrams per KC Bucket
- Cold-start Subsets: Strict ($f_{train} = 0$), k -shot subsets ($f_{train} \leq 5$ or ≤ 10) & Warm cohorts

5. Hygiene Audit & Reproducibility

7-Channel Leakage Checklist Audit (L1-L7)

Automatically Generated Reports → Standardized LaTeX/PDF Artifacts